8 Oscillators

FUNCTION GENERATOR

- A function generator is a device that is capable of producing different types of a waveform as the output signal.
- The function generator can produce sinusoidal, square, triangular, sawtooth signals etc., of variable frequencies.
- Oscillators that produce sinusoidal output are called linear oscillators, while the oscillators that produce square or triangular output are referred as non-linear oscillators.

Grey Matter Alert!

The clock signal used in a microprocessor is nothing but an oscillator.

OSCILLATOR

- The positive feedback amplifier which has loop gain >1 and satisfies the necessary phase condition, works as an oscillator.

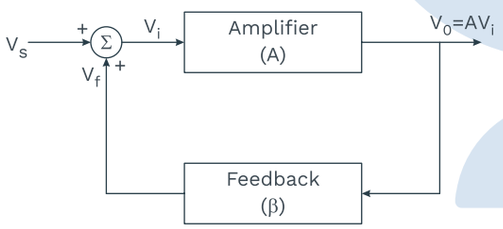


Fig. 8.1 Feedback circuit

- An oscillator producing sinusoidal output is called a harmonic oscillator. If an oscillator produces other than the sinusoidal output, then it is called a relaxation oscillator.

Applications:

1) Used in function generator.
2) Used in communication equipment.
3) Modulation and demodulation.
4) Clock in microprocessor.

Operation of oscillations:

If $V_i = 0$, then there are no oscillatios in the input signal. When input signal is non-zero then, $V_0 = AV_i$.

Feedback voltage = $\beta V_0 = \beta A V_i$

Loop gain = $A\beta$

Rack Your Brain

Why we cannot produce oscillations using negative feedback?

- For self-sustained oscillation, V_f can be made equal to V_i.
- Self-sustained implies that without any source, the oscillations can continue.

$$V_f = A\beta V_i = V_i$$

$$\therefore \quad A\beta = 1$$

- In such case, the source is initially connected to build up oscillations, and then the source can be disconnected once the oscillations have been set up.

$A\beta = 1$: Barkhausen criteria

Barkhausen criteria:

$$A\beta = 1$$

- Total phase shift around a loop must be a multiple of 2π.

$0, 2\pi, 4\pi, 6\pi, ...$

- The magnitude of the product of an amplifier gain and the feedback factor must be unity.

$$|A\beta| = 1$$

- If $A\beta = 1$, then input can be removed and oscillations become self-sustained.
- If $A\beta > 1$, then the signal used for feedback will be more than the input signal such that the oscillations keep on increasing.

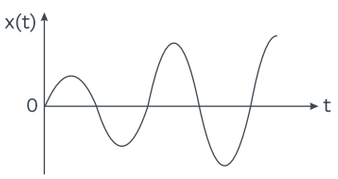


Fig. 8.2 Oscillations

- From Barkhausen criteria, the angle of $A\beta = 0°$ which means the signal used for feedback has the same phase as the input. So, if we use CE amplifier which has a phase shift of 180°, then the feedback network must provide a phase shift of 180° to make the total phase shift as 360°.

Grey Matter Alert!

High-frequency carrier signals in the communication system are produced by oscillators.

Rack Your Brain

What happens if only the magnitude condition is satisfied in the Barkhausen criteria? Will it be able to produce oscillations?

Previous Years' Question

Match the following:

List-1

a) Hartley
b) Wien-bridge
c) Crystal

List-II

1) Low frequency oscillator
2) High frequency oscillator
3) Stable frequency oscillator
4) Relaxation oscillator
5) Negative resistance oscillator

Types of oscillator:

1) **Audio frequency oscillator (20 Hz < f < 20 kHz)**
   - **i)** Wien-bridge oscillator
   - **ii)** RC phase shift oscillator

2) **Radio frequency oscillator (f > 20 kHz)**
   - **i)** Hartley oscillator
   - **ii)** Colpitts oscillator
   - **iii)** Clapp oscillator
   - **iv)** Crystal oscillator

Note: For the circuit to oscillate at a single frequency, Barkhausen criteria must be satisfied.

- The transfer function of the feedback network is-

$$\beta = \frac{V_f}{V_0} = \beta_{real} + j\beta_{imaginary}$$

- The gain of amplifier is always real (A = real) Barkhausen criteria: $A\beta = 1$. So, β must also be real.

  $\therefore \quad \beta_{imaginary} = 0 \Rightarrow$ Frequency of oscillations

- Using $A\beta_{real} = 1$, we can compute the required gain of an amplifier.

Example 1:
Why is the LC oscillator not used at low frequency?

Sol:

Frequency of oscillations, $f = \frac{1}{\sqrt{LC}}$

At low frequencies, the value of inductance required is high. So, the circuit would be bulky.

Example 2:
What is the difference between an amplifier and an oscillator?

Rack Your Brain

Can we use the Wien-bridge oscillator at microwave frequencies?

Grey Matter Alert!

Normal oscillator fails at microwave frequencies.

Sol:

An amplifier is an energy converter that takes energy from DC biasing and provides this energy to AC input in order to increase its amplitude. The oscillator needs an external energy source to start but does not need any source to maintain the continuous oscillations.

Wien-bridge oscillator:

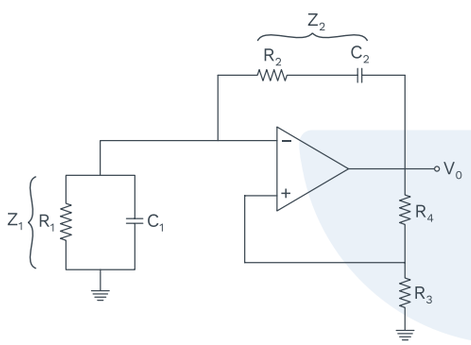


Fig. 8.3 Wien-bridge oscillator circuit

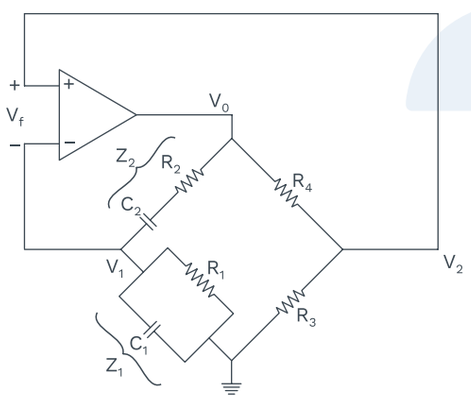


Fig. 8.4 Wien-bridge oscillator using four-arm bridge

$$V_1 = \frac{V_0 Z_1}{Z_1 + Z_2}$$

$$V_2 = \frac{V_0 R_3}{R_4 + R_3}$$

$$V_f = V_2 - V_1$$

$$V_f = V_0 \left[\frac{R_3}{R_4 + R_3} - \frac{Z_1}{Z_1 + Z_2} \right]$$

Rack Your Brain

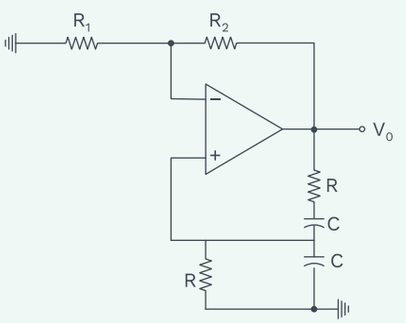
Why only inverting op-amp is used in oscillator circuits?

Grey Matter Alert!

The oscillator requires energy only at the starting of oscillation process.

Previous Years' Question

The configuration of given figure is a-



a) Precision integrator
b) Hartley oscillator
c) Butterworth high pass filter
d) Wien-bridge oscillator

If $V_f = V_i$, then circuit act as an oscillator.

$$\beta = \frac{V_f}{V_0} = \left[\frac{R_3}{R_4 + R_3} - \frac{Z_1}{Z_1 + Z_2}\right]$$

$$Z_1 = R_1 \,||\, \frac{1}{sC_1}$$

$$Z_1 = \frac{R_1}{1 + sC_1R_1}$$

$$Z_2 = R_2 + \frac{1}{sC_2}$$

$$Z_2 = \frac{1 + sC_2R_2}{sC_2}$$

$$\beta = \frac{R_3}{R_4 + R_3} - \frac{\frac{R_1}{(1 + sC_1R_1)}}{\left(\frac{R_1}{1 + sC_1R_1} + \frac{(1 + sC_2R_2)}{sC_2}\right)}$$

$$\beta = \frac{R_3}{R_4 + R_3} - \frac{sC_2R_1}{sC_2R_1 + (1 + sC_1R_1)(1 + sC_2R_2)}$$

$$\beta = \frac{R_3}{R_4 + R_3} - \frac{sC_2R_1}{sC_2R_1 + 1 + s^2R_1R_2C_1C_2 + s(R_1C_1 + R_2C_2)}$$

Put $s = j\omega$, we get-

$$\beta = \frac{R_3}{R_4 + R_3} - \frac{j\omega C_2R_1}{(1 - \omega^2R_1R_2C_1C_2) + j\omega(R_1C_1 + R_2C_2 + R_1C_2)}$$

For frequency of oscillations, β must be real.

$$\Rightarrow \; 1 - \omega^2R_1R_2C_1C_2 = 0$$

$$\omega = \frac{1}{\sqrt{R_1R_2C_1C_2}}$$

$$f = \frac{1}{2\pi\sqrt{R_1R_2C_1C_2}}$$

Rack Your Brain

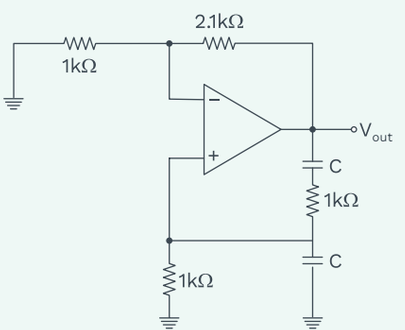
Can we use more than three RC circuits in the phase shift oscillator?

Grey Matter Alert!

Three RC networks are necessarily required in the phase shift oscillators.

Previous Years' Question

The value of C required for sinusoidal oscillation of frequency 1 kHz in the circuit of below figure is-



a) $\frac{1}{2\pi}\,\mu F$ **b)** $2\pi\,\mu F$

c) $\frac{1}{2\pi\sqrt{6}}\,\mu F$ **d)** $2\pi\sqrt{6}\,\mu F$

In such condition, we get-

$$\beta = \frac{R_3}{R_4+R_3} - \frac{R_1C_2}{R_1C_1+R_2C_2+R_1C_2}$$

For Barkhausen criteria-

$|A\beta| = 1$

$$|\beta| = \left|\frac{V_0}{V_f}\right| \Rightarrow |A| \to \infty \text{ then } \beta \to 0$$

$$\therefore \frac{R_3}{R_4+R_3} - \frac{R_1C_2}{R_1C_1+R_2C_2+R_1C_2} = 0$$

$$1+\frac{R_4}{R_3} = \frac{C_1}{C_2} + \frac{R_2}{R_1} + 1$$

$$\frac{R_4}{R_3} = \frac{R_2}{R_1} + \frac{C_1}{C_2}$$

Hence, two conditions that must be satisfied in Barkhausen criteria are-

a) $\omega = \dfrac{1}{\sqrt{R_1R_2C_1C_2}}$

b) $\dfrac{R_4}{R_3} = \dfrac{R_2}{R_1} + \dfrac{C_1}{C_2}$

Advantages:

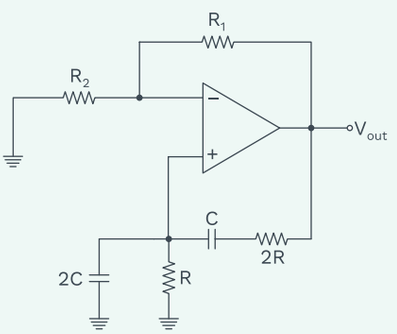
- Used as audio frequency oscillator.
- Used as a master oscillator in the signal generator.
- High-frequency stability due to bridge.

Drawbacks:

- A practical op-amp has a single break frequency and if this is used to design Wien bridge oscillator, then it suffers from RC phase lag. So, the RC phase lead must be designed to compensate the drawback of Wien-bridge oscillator.

Previous Years' Question

The circuit shown in the figure has an ideal Op-amp. The oscillation frequency and the condition to sustain the oscillations, respectively, are-



a) $\dfrac{1}{CR}$ and $R_1 = R_2$

b) $\dfrac{1}{CR}$ and $R_1 = 4R_2$

c) $\dfrac{1}{2CR}$ and $R_1 = R_2$

d) $\dfrac{1}{2CR}$ and $R_1 = 4R_2$

Rack Your Brain

Why common emitter configuration of BJT is used in an oscillator circuit?

RC phase-shift oscillator:

a) Using op-amp:

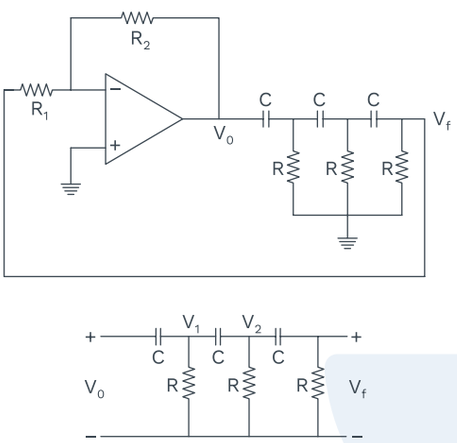


Fig. 8.5 RC phase shift oscillator using op-amp

- It is used to produce sustained well shaped sinusoidal oscillation.
- It is used in different applications such as local oscillators for the synchronous receiver, musical instruments, study purpose etc.
- If we observe amplifier as negative feedback, it provides 180° phase shift, and hence in order to realize an oscillator, the feedback network should also provide a phase shift of 180°. Therefore, it is called as RC phase shift oscillator.

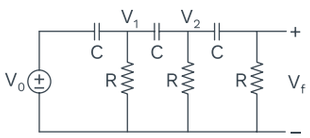


Fig. 8.6 RC circuits

$$V_f = V_2 \times \frac{R}{R + \frac{1}{sC}} = V_2\left(\frac{sCR}{1+sCR}\right)$$

$$V_2 = V_f \frac{(1+sCR)}{sCR} \qquad ...(i)$$

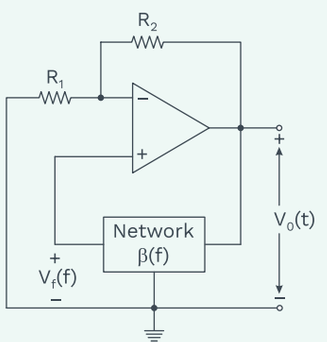
Applying KCL at node V_2, we get -

Previous Years' Question

The circuit in the figure employs positive feedback and is intended to generate sinusoidal oscillation. If at a frequency f_0,

$$\beta(f) = \frac{V_f(f)}{V_0(f)} = \frac{1}{6}\angle 0°$$

then to sustain oscillation at the frequency, which condition is satisfied?



a) $R_2 = 5R_1$ **b)** $R_2 = 6R_1$

c) $R_2 = \frac{R_1}{6}$ **d)** $R_2 = \frac{R_1}{5}$

$$\frac{V_2-V_1}{1/sC}+\frac{V_2}{R}+\frac{V_2}{R+\frac{1}{sC}}=0$$

$$sC(V_2-V_1)+\frac{V_2}{R}+\frac{sC\,V_2}{1+sCR}=0$$

$$V_2\left[sC+\frac{1}{R}+\frac{sC}{1+sCR}\right]=sCV_1$$

$$V_1=\left[1+\frac{1}{sCR}+\frac{1}{1+sCR}\right]V_2 \qquad ...(ii)$$

Applying KCL at node V_1, we get-

$$\frac{V_1-V_0}{1/sC}+\frac{V_1}{R}+\frac{V_1-V_2}{1/sC}=0$$

$$\left(2sC+\frac{1}{R}\right)V_1=sCV_0+sCV_2$$

Putting the value of V_1 from equation (ii), we get-

$$\left(\frac{1+2sCR}{R}\right)\left[1+\frac{1}{sCR}+\frac{1}{1+sCR}\right]V_2-sCV_2=sCV_0$$

$$\left[\left(\frac{1+2sCR}{sCR}\right)\left(\frac{3(sCR)+(sCR)^2+1}{sCR(1+sCR)}\right)-1\right]V_2=V_0$$

$$\left[\frac{2(sCR)^3+7(sCR)^2+5(sCR)+1-(sCR)^3-(sCR)^2}{(sCR)^2(1+sCR)}\right]V_2=V_0$$

Putting the value of V_2 from equation (i), we get-

$$\left[\frac{(sCR)^3+6(sCR)^2+5(sCR)+1}{(sCR)^2(1+sCR)}\right]\cdot\frac{(1+sCR)}{sCR}\cdot V_f=V_0$$

$$\left[\frac{(sCR)^3+6(sCR)^2+5(sCR)+1}{(sCR)^3}\right]V_f=V_0$$

$$\beta=\frac{V_f}{V_0}=\frac{(sCR)^3}{(sCR)^3+6(sCR)^2+(5sCR)+1}$$

$$\beta=\frac{1}{1+\frac{6}{sCR}+\frac{5}{(sCR)^2}+\frac{1}{(sCR)^3}}$$

Rack Your Brain

What is the frequency of oscillations of phase shift oscillator?

Grey Matter Alert!

Op-amp, BJT or FET, are used to construct phase shift oscillators.

$$\beta(j\omega) = \frac{1}{1 + \frac{6}{j\omega RC} + \frac{5}{-\omega^2R^2C^2} - \frac{1}{j(\omega RC)^3}}$$

For frequency of oscillations, $\beta_{imag} = 0$

$$\frac{6}{\omega RC} = \frac{1}{(\omega RC)^3}$$

$$(\omega RC)^2 = \frac{1}{6}$$

$$\omega = \frac{1}{RC\sqrt{6}}$$

$$\beta_{real} = \frac{1}{1 - \frac{5}{\omega^2R^2C^2}}$$

$$\beta_{real} = \frac{1}{1 - 30}$$

$$\beta_{real} = \frac{1}{-29} (180° \text{ phase shift})$$

For Barkhausen criteria, $|A\beta| = 1$

$$|A| = \frac{1}{|\beta|} = 29 = \frac{R_f}{R_1}$$

$$R_f = 29R_1$$

- To avoid the loading effect, we can connect a buffer between the stage of the RC feedback network and then we can simply take the product of the transfer function of each stage of the RC network.

b) Using BJT:

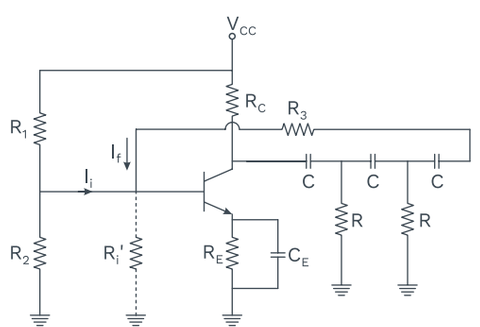


Fig. 8.7 a) RC phase shift oscillator using BJT

Rack Your Brain

Why is the tank circuit important in the LC oscillator?

Grey Matter Alert!

The tank circuit is nothing but an LC oscillator circuit.

- Using BJT, we have designed a common emitter amplifier whose Q-point must lie in the active region.
- Q-point is set by R_1, R_2, R_C, R_E.
- The three stages of the RC feedback network are identical.

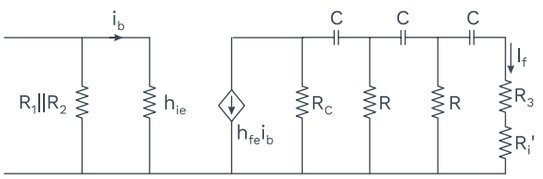


Fig. 8.7 b) BJT oscillator model circuit

Effective input resistance:

$$R_i' = h_{ie} \parallel R_{Th} = (h_{ie} \parallel R_1 \parallel R_2)$$

Frequency of oscillation, $f_{oscillation} = \dfrac{1}{2\pi RC\sqrt{4K+6}}$

where, $K = \dfrac{R_C}{R}$

Condition for oscillation-

$$h_{fe} > 4K + 23 + \frac{29}{K}$$

Differentiating the above equation with respect to K, we get-

$$\frac{dh_{fe}}{dK} = \frac{4K^2 - 29}{K^2} = 0$$

$$K = \frac{\sqrt{29}}{2}$$

$\therefore$ $h_{fe}(min) = 44.54$

Rack Your Brain

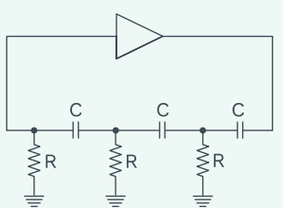
What is the Hartley oscillator?

Grey Matter Alert!

Inverting amplifier is generally used to provide 180° phase shift in an oscillator.

Previous Years' Question

The oscillator circuit shown in the figure is an ideal inverting amplifier, its frequency of oscillation (in Hz) is-



a) $\dfrac{1}{(2\pi\sqrt{6}RC)}$ **b)** $\dfrac{1}{(2\pi RC)}$

c) $\dfrac{1}{(\sqrt{6}RC)}$ **d)** $\dfrac{\sqrt{6}}{(2\pi RC)}$

c) Using FET:

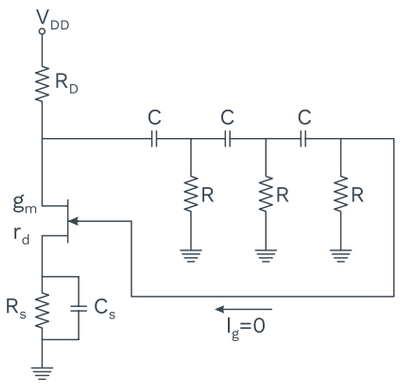


Fig. 8.7 c) RC oscillator using FET

- Similar to the op-amp, the input current in FET is zero, so the frequency of an oscillator is the same as that of an op-amp circuit.
- For FET, we replaced the common emitter amplifier of BJT by the common source amplifier of FET.

$$f_{oscillation} = \frac{1}{2\pi RC\sqrt{6}}$$

LC oscillator:

- LC oscillator makes use of the tank circuit in order to generate a sinusoidal wave of high frequency.
- LC oscillator is used to generate frequencies in the order of 500 MHz.

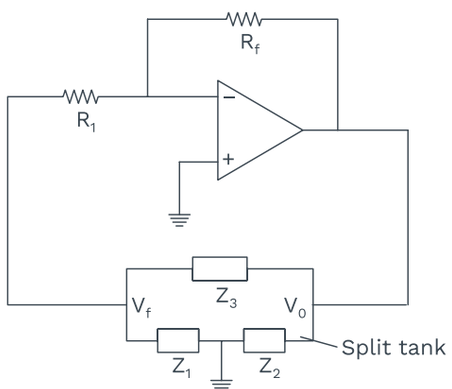


Fig. 8.8 LC oscillator

Grey Matter Alert!

The difference between Hartley and Colpitts oscillators is the number of inductors and capacitors used.

Rack Your Brain

Which types of oscillators are used at microwave frequencies?

Phase splitter:

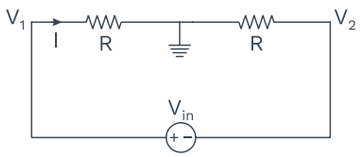


Fig. 8.9 a) Phase splitter circuit using R

$$I = \frac{V_1 - 0}{R} = \frac{0 - V_2}{R}$$

$\Rightarrow V_2 = -V_1$

So, the magnitude of both voltages is equal, but they have a phase shift of 180°.

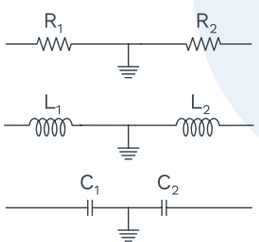


Fig. 8.9 b) Phase splitter circuit using R, L, C

- **Gain of inverting amplifier:**

$$G = \frac{-R_f}{R_1}$$

$$I = \frac{V_0 - 0}{Z_2} = \frac{0 - V_f}{Z_1}$$

Feedback factor, $\beta = \frac{V_f}{V_0} = \frac{-Z_1}{Z_2}$

From Barkhausen criteria, $A = \frac{1}{\beta}$

$$\frac{-R_f}{R_1} = \frac{-Z_2}{Z_1}$$

$$\Rightarrow \frac{R_f}{R_1} = \frac{Z_2}{Z_1}$$

Since current would exist in a loop,

$$I(Z_2 + Z_1 + Z_3) = 0$$

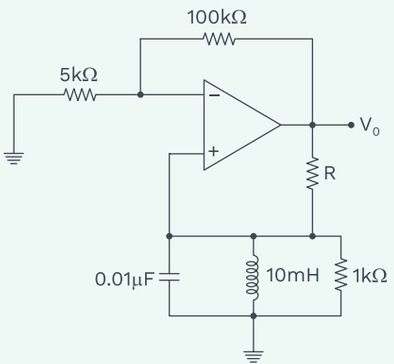
$$Z_1 + Z_2 + Z_3 = 0$$

Grey Matter Alert!

BJT is always used in the common-emitter configuration to provide 180° phase shift between input and output.

Previous Years' Question

The value of R of the oscillator is shown in the given figure, so that it just oscillates at an angular frequency of 'ω'. The value of 'ω' and the required value of R will be, respectively-



a) 10^5 rad/sec, $2\times10^4\ \Omega$
b) 2×10^4 rad/sec, $2\times10^4\ \Omega$
c) 2×10^4 rad/sec, $10^5\ \Omega$
d) 10^5 rad/sec, $10^5\ \Omega$

Hartley oscillator:

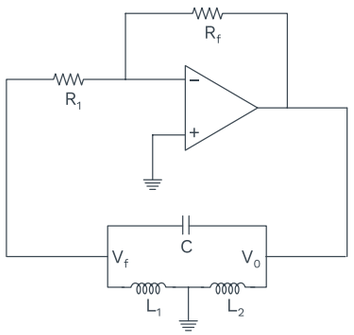


Fig. 8.10 a) Hartley oscillator circuit using op-amp

- 2 inductors + 1 capacitor
- To find frequency of oscillation,

$$Z_1 + Z_2 + Z_3 = 0$$

$$j\omega L_1 + j\omega L_2 - \frac{j}{\omega C} = 0$$

$$\omega = \frac{1}{\sqrt{(L_1 + L_2)C}}$$

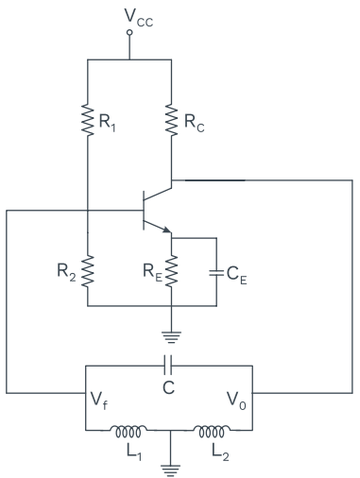


Fig. 8.10 b) Hartley oscillator using BJT

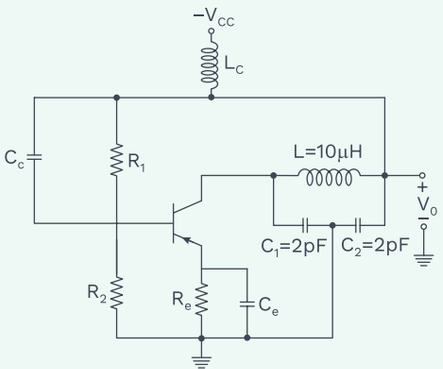
- The amplitude of output remains the same over the working frequency range of op-amp and the main disadvantage of this oscillator is its high harmonic content, due to which it is not suitable for generating pure sine waves.

Rack Your Brain

Which oscillator is used in the modulation and the demodulation of message signals?

Previous Years' Question

The oscillator circuit shown in the figure is-



a) Hartley oscillator with $f_{oscillation}$ = 79.6 MHz.

b) Colpitts oscillator with $f_{oscillation}$ = 50.3 MHz.

c) Hartley oscillator with $f_{oscillation}$ = 159.2 MHz.

d) Colpitts oscillator with $f_{oscillation}$ = 159.2 MHz.

- In the case of BJT, we replace inverting amplifier by a common emitter amplifier (180° phase shift).
- Gain $= \frac{-R_C}{r_e}$

Colpitts oscillator:

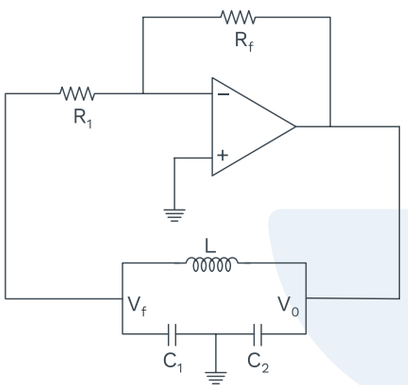


Fig. 8.11) a) Colpitts oscillator

Frequency of oscillation,

$$Z_1 + Z_2 + Z_3 = 0$$

$$\frac{-j}{\omega C_1} + \frac{-j}{\omega C_2} + j\omega L = 0$$

$$\omega L = \frac{1}{\omega}\left(\frac{1}{C_1} + \frac{1}{C_2}\right)$$

$$\omega = \frac{1}{\sqrt{L \cdot \frac{C_1 C_2}{C_1 + C_2}}}$$

$$f_{oscillation} = \frac{1}{2\pi}\sqrt{\frac{1}{L}\left(\frac{1}{C_1} + \frac{1}{C_2}\right)}$$

$$\frac{R_f}{R_1} = \frac{Z_2}{Z_1} = \frac{C_1}{C_2}$$

Colpitts oscillator using BJT:

The main advantage of the Colpitts oscillator is improved performance at the high frequencies due to the low reactance path provided by the capacitor at the high frequency.

Grey Matter Alert!

Crystal oscillators produce highly stable waveforms mainly due to the quartz crystal.

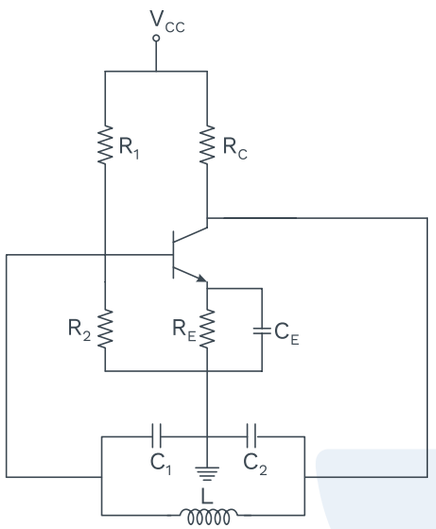


Fig. 8.11 b) Colpitts oscillator using BJT

- A Colpitts oscillator is used at a high frequency.

Clapp oscillator:

- In the Clapp oscillator circuit, an additional capacitor is added in series with the inductor.

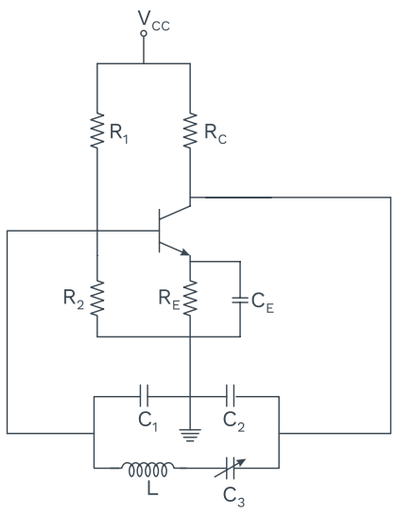


Fig. 8.12 Clapp oscillator

- The main purpose of adding capacitor C_3 is to improve the frequency stability of the system.

Grey Matter Alert!

A crystal oscillator is generally used to provide a clock signal in the digital circuits.

Rack Your Brain

Why crystal oscillator has a very-high quality factor?

- $j\omega L - \frac{j}{\omega C_3} - \frac{j}{\omega C_1} - \frac{j}{\omega C_2} = 0$

  $$\omega = \sqrt{\frac{1}{L}\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)}$$

- Usually, capacitor C_3 dominates over other two capacitors, then we get -

  $$\omega = \sqrt{\frac{1}{LC_3}}$$

  This ensures the phase splitter circuit does not decide the frequency.

Crystal oscillator:

- An electrical resonant circuit is replaced by mechanical vibrating crystal. The crystal has a high degree of stability for maintaining frequency constant.
- In the crystal oscillator, we use piezoelectric material such as quartz which can produce electric voltage when it is put under mechanical strain.

AC equivalent of crystal oscillator:

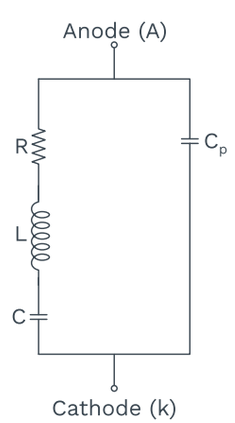


Fig. 8.13 a) AC equivalent of crystal oscillator

R = Viscous damping or internal losses
L = Mass of crystal
C = Motional capacitance
C_p = Capacitance between anode and cathode

Rack Your Brain

How is the Hartley oscillator different from Colpitts oscillator?

- Crystal oscillator has two oscillation frequencies, one due to the series capacitor and the other due to the parallel capacitor.
- The first has low frequency and the impedance is low, and the other has high frequency and impedance is high.

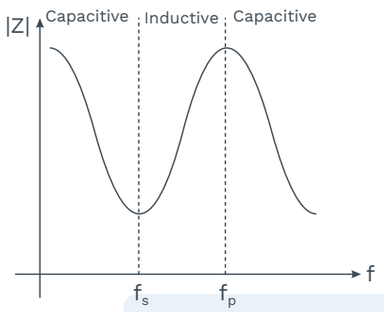


Fig. 8.13 b) Z vs f for crystal oscillator

For series resonance,

$$f_s = \frac{1}{2\pi\sqrt{LC}}$$

For parallel resonance,

$$f_p = \frac{1}{2\pi\sqrt{LC_{eq}}}$$

$$C_{eq} = C + C_p$$

- Oscillation frequency depends on the size of the crystal.

$$f_0 \propto \frac{1}{\text{thickness}}$$

- It has high quality factor (5000–25,000).

Example 3:

For the Wien-bridge oscillator shown below, determine the oscillation frequency if $R_1 = R_2 = 50\ k\Omega$, $C_1 = C_2 = 0.001\ \mu F$, $R_3 = 400\ k\Omega$ and $R_4 = 100\ k\Omega$.

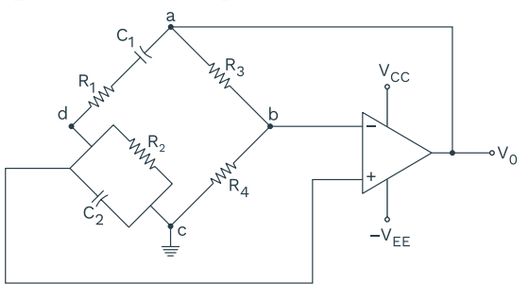


Fig. 8.14

a) 5 kHz **b)** 3.18 kHz **c)** 31.2 kHz **d)** 10 kHz

Sol:

Given, $R_1 = R_2 = 50\ k\Omega$

$C_1 = C_2 = 0.001\ \mu F$

$R_3 = 400\ k\Omega$ $R_4 = 100\ k\Omega$

In the Wien-bridge oscillator, conditions must be satisfied in Barkhausen criteria.

$$\frac{R_3}{R_4} = \frac{400}{100} = 4$$

The ratio of $R_3/R_4 > 2$.
It will make the circuit oscillate at the oscillation frequency.

We know that the oscillation frequency for Wien-bridge oscillator is-

$$f_0 = \frac{1}{2\pi\sqrt{R_1C_1R_2C_2}}$$

$$f_0 = \frac{1}{2\pi\sqrt{50\times10^3\times0.001\times10^{-6}\times50\times10^3\times0.001\times10^{-6}}}$$

$$= \frac{1}{2\pi\times50\times10^3\times0.001\times10^{-6}} = 3.18\,kHz$$

Hence, the correct option is **b)**.

Example 4:

What is the feedback factor required for the RC phase shift oscillator to behave like an oscillator?

a) –29 **b)** $-\frac{1}{29}$ **c)** 29 **d)** $\frac{1}{29}$

Sol:

RC phase shift oscillator is shown below.

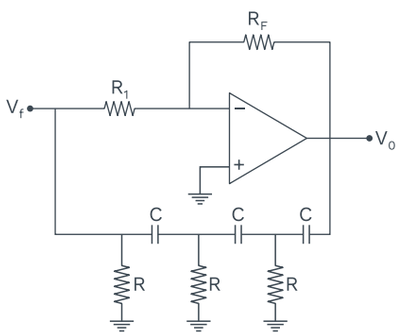


Fig. 8.15

Feedback factor is $\beta = \frac{V_f}{V_0}$

For oscillations, feedback factor is $\frac{V_f}{V_0} = -\frac{1}{29}$

Hence, the correct option is **b)**.

Example 5:
For the Wien-bridge oscillator shown below, find the value of R and C such that it oscillates at 1 kHz.

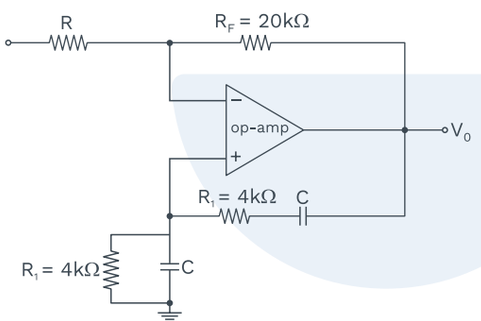


Fig. 8.16

a) R = 20 kΩ, C = 0.04 μF

b) R < 10 kΩ, C = 0.0397 μF

c) R > 10 kΩ, C = 0.04 μF

d) R = 10 kΩ, C = 0.02 μF

Sol:
Given, f_0 = 1 kHz

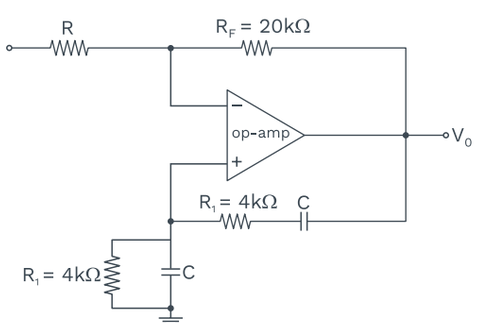


Fig. 8.17

For the Wien-bridge oscillator, condition of gain for sustained oscillation is-

$A_V > 3$

But $A_V = 1 + \frac{R_F}{R}$

$$\left(1+\frac{R_F}{R}\right) > 3$$

$$\left(1+\frac{20\,k\Omega}{R}\right) > 3$$

$$\frac{20\,k\Omega}{R} > 2$$

$$R < 10\ k\Omega$$

We know that,

Frequency of oscillation, $f_0 = \frac{1}{2\pi R_1 C}$

$$C = \frac{1}{2\pi R_1 f_0} = \frac{1}{2 \times \pi \times 4 \times 10^3 \times 1 \times 10^3}$$

$$C = 0.0397\ \mu F$$

Hence, the correct option is **b)**.

Example 6:

An oscillator circuit is shown in the figure; what is the frequency of oscillation in kHz?

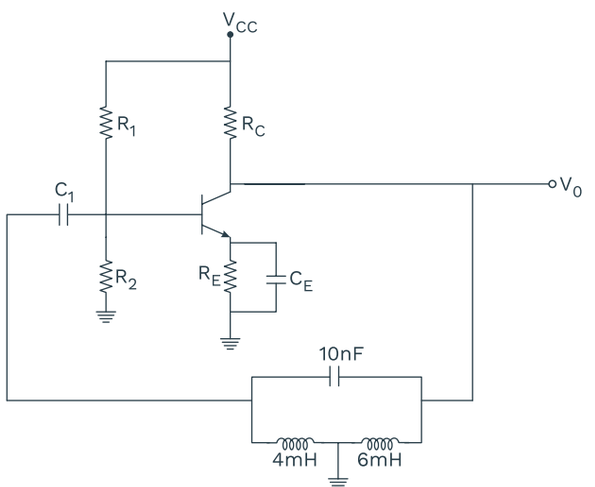


Fig. 8.18

Sol:
The given oscillator is a Hartley oscillator.
The frequency of oscillation of the Hartley oscillator is given by-

$$f = \frac{1}{2\pi\sqrt{(L_1 + L_2)C}}$$

$L_1 = 4mH$

$L_2 = 6mH$

$C = 10nF$

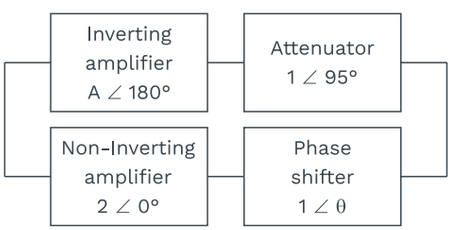
$$f = \frac{1}{2\pi\sqrt{10 \times 10^{-3} \times 10 \times 10^{-9}}}$$

$$f = \frac{10^5}{2\pi}$$

Frequency, f = 15.9 kHz.

Hence, the correct answer is 15.9.

Example 7:
The closed-loop positive feedback system is shown below with all the elements along their magnitude and phase. Find the phase angle 'θ' of the phase shift oscillator block in degrees?



Sol:
For closed-loop system to behave as an oscillator-
Total phase angle = 360°

$\therefore$ $180° + 95° + \theta = 360°$

$\theta = 360° - 180° - 95° = 85°$

$\theta = 85°$

Hence, the correct answer is 85.

Chapter Summary



- A waveform can be generated by using a device known as a function generator.
- A linear oscillator produces sinusoidal output, whereas a nonlinear oscillator produces a square or triangular waveform.
- Oscillators have multiple applications like function generators, communication systems, clock generators etc.
- The product of forward path gain 'A' and feedback factor 'β' is known as loop gain, and it is given by Aβ.
- For a closed-loop system to work as an oscillator, it must satisfy Barkhausen criteria, i.e., magnitude criteria and phase criteria.
  - **i)** $|A\beta| = 1$
  - **ii)** Total phase change around a closed loop must be a multiple of 2π
- For a closed-loop system to oscillate at a single frequency, the Barkhausen criteria must be satisfied at a single frequency.
- Audio frequency oscillator produces waveform in the frequency range of 20 Hz–20 kHz such as-
  - **i)** Wien-bridge oscillator
  - **ii)** RC phase shift oscillator
- A radio frequency oscillator produces waveform in a frequency range greater than 20kHz such as-
  - **i)** Hartley oscillator
  - **ii)** Colpitts oscillator
  - **iii)** Clapp oscillator
  - **iv)** Crystal oscillator
- The frequency of oscillation of the Wien-bridge oscillator is-

$$f = \frac{1}{2\pi\sqrt{R_1R_2C_1C_2}}$$

- RC phase shift oscillator can be built by using op-amp, BJT or FET.
- In the RC phase shift oscillator, three RC circuits are required.
- The frequency of oscillation of the RC phase shift oscillator is-

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

Feedback factor, $\beta = -\frac{1}{29}$

- LC oscillator is constructed by using a tank circuit.
- The frequency of oscillation of the Hartley oscillator is-

$$f = \frac{1}{2\pi\sqrt{(L_1 + L_2)C}}$$

- The frequency of oscillation of the Colpitts oscillator is-

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{L}\left(\frac{1}{C_1} + \frac{1}{C_2}\right)}$$

- Oscillations produced by crystal oscillators are highly stable, which uses piezo material such as quartz. It produces electrical voltage when it is put under mechanical strain.
- Crystal oscillator has a very high-quality factor.

PRACTICE QUESTIONS

Q1 Hartley oscillator is a-

a) High frequency oscillator
b) Low frequency oscillator
c) Stable frequency oscillator
d) Relaxation oscillator

Q2 Wien-bridge oscillator is a-

a) High frequency oscillator
b) Low frequency oscillator
c) Stable frequency oscillator
d) Relaxation oscillator

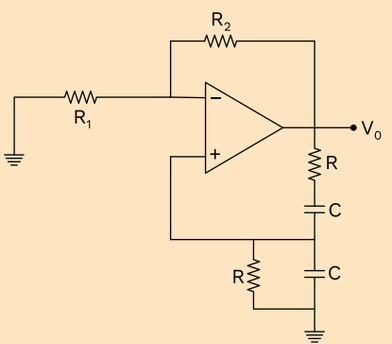
Q3 The range of the audio frequency oscillator is-

a) 1 Hz–1 kHz
b) 20 Hz–20 MHz
c) 20 Hz–20 kHz
d) 1 kHz–1 MHz

Q4 LC oscillator is not used at low frequency mainly because of-

a) Capacitor
b) Inductor
c) Inductor and capacitor
d) None of the above

Q5 An oscillator circuit is shown below. What is the frequency of oscillation?



a) $\frac{1}{RC}$
b) $\frac{1}{\pi RC}$
c) $\frac{1}{2\pi RC\sqrt{6}}$
d) $\frac{1}{2\pi RC}$

Q6 Crystal oscillator generates-

a) Triangular wave
b) Sawtooth wave
c) Both
d) None of these

Q7 What is the relation between feedback resistance (R_F) and resistance (R_1) in the case of a phase-shift oscillator?

a) $R_F = R_1/29$
b) $R_F = 29R_1$
c) $R_1 = 29R_F$
d) $R_1 = R_F$

Q8 Colpitts oscillator uses-

a) 2 inductors and 1 capacitor
b) 1 inductor and 1 capacitor
c) 1 inductor and 2 capacitors
d) 2 inductors and 2 capacitors

Q9 Crystal oscillator produces-

a) Highly unstable wave
b) Marginally stable wave
c) Highly stable wave
d) None of the above

Q10 Range of oscillation frequency 1 Hz–10 MHz is of -

a) Wien-bridge oscillator
b) Phase shift oscillator
c) Colpitts oscillator
d) Clapp oscillator

Answer Key:

1) a)	2) b)	3) c)	4) b)
5) c)	6) c)	7) b)	8) c)
9) c)	10) b)		